1. Introduction and Background

1.1 Project Background and Significance

Rice University's Office of Information Technology (OIT), in collaboration with the Office of Institutional Effectiveness (OIE), oversees an annual outcomes assessment process in which every academic degree program submits documentation describing how well it is achieving its Program Learning Outcomes (PLOs). These submissions, known as Rice Outcomes Assessment Reviews (ROARs), require faculty to describe assessment methods, report results, and outline follow-up actions intended to improve student learning.

Each year, OIT receives a large volume of ROAR submissions spanning dozens of programs. The review of these submissions is currently a manual process. Staff must read each ROAR to verify that the methods align with the stated PLOs, that results correspond to the methods described, and that the analysis and follow-up actions are specific and meaningful. Because of the scale and complexity of the submissions, this review process often extends over long timelines, which delays feedback and hinders the whole process.

Timely feedback is critical to the effectiveness of outcomes assessment. When feedback is delayed, programs are less likely to act on the results. Faster feedback enables programs to identify issues sooner, make instructional or curricular adjustments more quickly, and ultimately improve educational outcomes for students.

The goal of this project is to support OIT and Institutional Effectiveness by developing a pre-screening software system that provides an automated "first set of eyes" on ROAR submissions. The proposed system is intended to flag major structural or semantic issues so that reviewers can focus their attention where it is most needed. The system has the potential to reduce review time and improve the overall effectiveness of the outcomes assessment process.

In the longer term, such a tool could be extended beyond internal screening to help academic programs self-evaluate submissions before they are formally submitted, or to support similar assessment workflows in non-academic units. As a result, this project has the potential to create a lasting impact by both improving operational efficiency and helping drive continuous improvement in student learning.

1.2 Project Objectives

Objective 1: Develop a structured data extraction pipeline for ROAR submissions.

- Create a system capable of parsing ROAR documents: identifying and extracting key fields such as Program Learning Outcomes, assessment methods, reported results, analysis, and follow-up actions.

Objective 2: Design automated checks for ROAR completeness and compliance.

- Implement rule-based or model-based methods to detect missing fields, formatting inconsistencies, duplicated content, and/or other structural issues that can be identified without deep semantic interpretation or more complicated models.

Objective 3: Identify PLO and degree program correlation of poor ROAR submissions.

- Analyze data to determine which program characteristics, PLO features, or other elements are associated with lower-quality submissions.

Objective 4: Build a semantic evaluation model to assess alignment and coherence within submissions.

- Develop a model that evaluates whether assessment methods align with stated PLOs and whether reported results correspond to the described methods.

Objective 5 (mentioned in the sponsor meeting as a potential feature): Implement a flagging system to identify potentially low-quality or inconsistent submissions.

- Create a system that generates interpretable flags indicating areas of concern (vague analysis, misaligned assessment, missing performance targets, etc).

Objective 6: Establish a validation framework using sponsor-provided test data.

- Define evaluation criteria and test the system on mock and historical ROAR submissions to assess whether the tool reliably identifies major structural and semantic issues.

Objective 7: Propose an output format suitable for practical use by OIE reviewers.

- Design a clear output (e.g., summary report or categorized flags) that supports rapid review and can be integrated into future workflows and department workflow improvements.

1.3 Literature Review and Related Work

Related technical approaches

Two technical areas are especially relevant to this project. The first is document information extraction, where NLP methods are used to convert unstructured reports into structured fields that can be analyzed automatically. Similar approaches have been applied in domains like clinical reporting, where automated extraction helps reduce manual review workload across large volumes of documents. These systems show that extracting key entities and sections from reports is feasible and provides a foundation for downstream evaluation, which is very relevant to the ROARS project.

The second area is rubric-aligned automated evaluation. Research in automated essay scoring and educational NLP shows that systems are more useful when they provide feedback across multiple criteria rather than assigning a single overall score. Trait-based evaluation improves interpretability and allows reviewers to understand why an item was flagged. Related work also demonstrates that NLP methods can

measure semantic similarity between learning outcomes and supporting text, suggesting that alignment between PLO statements and assessment descriptions can be evaluated computationally.

Limitations in prior work

Existing research generally focuses on student writing, grading systems, or generic information extraction. There is less work aimed at automated screening of institutional assessment reports as a workflow support tool. Many approaches prioritize classification or scoring rather than evaluating coherence across multiple report sections.

How this project builds on prior work

This project combines document extraction and rubric-based evaluation into a single pipeline tailored to ROAR submissions. First, the system will extract structured fields such as PLOs, assessment methods, results, analysis, and follow-up actions. Second, rule-based and model-based methods will evaluate alignment across these fields and generate interpretable flags. The project differs from prior work by focusing on support for institutional reviewers.

Sources

- Illinois State University Teaching & Learning Evaluation Center – Assessment Report Guidance https://tec.illinoisstate.edu/downloads/2024_TEC_Assessment_Report.pdf
- NLP-based information extraction from clinical reports (example of structured extraction workflows) https://pmc.ncbi.nlm.nih.gov/articles/PMC9055472/
- Automated essay scoring and rubric-based evaluation in education https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2020.572367/full
- Rubric-aligned evaluation using language models (trait-level assessment) https://arxiv.org/html/2602.04604v1
- NLP for mapping learning outcomes across educational levels https://dl.acm.org/doi/abs/10.1007/s10639-023-11877-4

2. Data Description and Evaluation Plan

2.1 Data Sources

The datasets used in this project originate from Rice University's Institutional Effectiveness (IE) team within the Institutional Data, Evaluation, Analytics, and Strategy (IDEAS) Office, operating under the Office of the Provost. The IE team oversees institutional accreditation and annual outcomes assessment processes, including the collection and review of Rice Outcomes Assessment Reports (ROARs).

The primary data consists of Program Learning Outcomes (PLOs) and corresponding ROAR submissions from 150+ academic degree programs across Rice. Each program defines formal PLOs describing what students should know or be able to do upon graduation. Programs then submit annual ROAR documents describing how those outcomes were assessed, what results were observed, and what improvement actions are planned. These submissions are part of Rice's compliance with accreditation standards established by the Southern Association of Colleges and Schools Commission on Colleges (SACSCOC).

The data is internally generated through Rice's established institutional assessment workflow. Academic programs submit ROARs annually through official university reporting channels, and the IDEAS Office stores and manages these documents. The project team will receive access to this dataset through secure Rice-managed platforms (Rice Box), in accordance with university data governance policies.

The dataset spans multiple annual reporting cycles. Because PLOs are assessed on a three-year rotation, the data allows for both multi-program and multi-year analysis. The geographic scope is institution-specific and confined to Rice in Houston, Texas, though it encompasses all academic divisions and disciplines (STEM, humanities, social sciences, and professional programs).

In terms of format, the data is primarily unstructured narrative text contained in Word documents or PDFs. Within these documents, there are embedded quantitative elements such as percentages, rubric scores, tables, and performance thresholds. Thus, the dataset is best characterized as semi-structured text with embedded numerical information.

The sponsor proposal document serves as the primary documentation describing the structure, purpose, and governance of the ROAR and PLO datasets. All data is internal institutional documents and will be handled confidentially. No student-identifiable information is included, and program-level assessment results and improvement plans are considered sensitive and will be anonymized in any public-facing outputs.

2.2 Dataset Structure

The primary unit of observation in this project is a PLO-ROAR pair, meaning each observation represents a single Program Learning Outcome (PLO) evaluated within a specific ROAR submission for a given program and reporting year. This structure allows the analysis to occur at the outcome level rather than at the full-document level, which is critical for assessing semantic alignment between stated learning objectives and reported assessment evidence. Across 150+ academic programs, there are currently 1,365 distinct PLOs. Because PLOs are assessed on a rotating three-year cycle and ROARs are submitted annually, the total number of modeling observations will depend on how many reporting years are included and how many PLOs are assessed in each cycle. As additional years are incorporated, the number of PLO-ROAR pairs is expected to reach the low-to-mid thousands.

Right now, the ROAR documents exist as full PDF or Word files and have not yet been parsed into structured fields. Each document contains narrative descriptions of assessment methods, reported results,

analysis of findings, and follow-up improvement actions. These materials are primarily unstructured text, though they frequently include embedded quantitative elements such as percentages, rubric scores, and performance thresholds presented within tables or visualizations. As a result, the dataset is best characterized as semi-structured: largely textual but containing numerical information that must be extracted through preprocessing.

In addition to the textual content, the dataset includes structured metadata associated with each ROAR submission. These metadata fields include program name, department, school, reporting year, degree level (undergraduate or graduate), and measure indicators (direct versus indirect assessment methods). These variables are categorical or time-based and will serve as structured features for exploratory analysis and modeling. The reporting year provides a temporal dimension that enables longitudinal analysis of assessment practices over time.

A labeled subset of ROAR submissions is available for supervised learning tasks. Currently, 39 ROARs have been manually flagged by Institutional Effectiveness experts, including 13 identified as unacceptable and 26 identified as acceptable. These labels are categorical and represent overall quality judgments. Additional ROAR documents are available but have not yet been flagged. Because the labeled subset is relatively small compared to the full corpus, model development will require careful validation procedures to avoid overfitting and class imbalance.

Overall, the dataset is multimodal, consisting of unstructured text (PLO statements and ROAR narrative sections), embedded numerical data (assessment results and performance indicators), categorical metadata (program-level and submission-level descriptors), and time-based variables (reporting year). This structure supports multiple levels of analysis, including document-level quality classification, PLO-level semantic alignment modeling, and cross-program or longitudinal trend analysis.

2.3 Data Quality and Limitations

Several limitations arise from how the data was generated and how it will be used in modeling.

First, the labeled dataset is small (39 flagged ROARs) relative to the overall corpus. This creates a high risk of overfitting in supervised learning models, particularly for text-based classifiers with large parameter spaces. With limited examples of unacceptable submissions, models may learn patterns that are overly specific to those particular documents rather than generalizable indicators of poor alignment or quality. Careful validation strategies will be necessary to minimize this risk.

Second, the quality labels themselves are based on expert judgment. While these evaluations are authoritative, they may reflect implicit standards, disciplinary expectations, or reviewer-specific norms that are not explicitly encoded in the documents. As a result, there is potential for subjectivity in the ground truth. Models trained on these labels may replicate historical review patterns, including any inconsistencies across reviewers or years. This introduces the possibility of algorithmic bias, particularly

if certain writing styles, disciplinary vocabularies, or reporting conventions correlate with labeled outcomes.

Third, incomplete or ambiguous reporting within ROARs presents interpretive challenges. Some submissions may lack explicit performance benchmarks, clearly defined improvement plans, or consistent terminology. While these omissions are substantively meaningful for evaluation, they may also make it difficult for automated systems to distinguish between genuinely low-quality assessments and concise but adequate reporting. The distinction between brevity and insufficiency may not always be clear in text alone.

From an ethical perspective, the data contains sensitive program-level assessment results and improvement plans. Although no student-identifiable information is present, automated flagging could influence perceptions of departmental performance if misinterpreted. For this reason, the system is explicitly designed as a decision-support tool that provides preliminary flags rather than final judgments. Transparency and interpretability in model outputs will be essential to ensure that reviewers understand why a submission was flagged.

Finally, temporal dynamics may affect consistency. Because PLOs are assessed on a rotating three-year cycle, different outcomes may be emphasized in different years, potentially resulting in uneven representation over time. Changes in institutional guidance or reviewer expectations over time might also shift what constitutes an “acceptable” submission, complicating longitudinal modeling.

To summarize, the primary limitations of this dataset relate to label scarcity, potential subjectivity, institutional specificity, and the risk of bias amplification. These considerations will directly inform model selection, validation design, and deployment strategy to ensure that the resulting system supports institutional effectiveness without introducing unintended distortions.

2.4 Planned Exploratory Data Analysis (EDA)

The first stage of EDA will focus on descriptive structural analysis. We will compute summary statistics on document length (word counts, sentence counts, section lengths) for both acceptable and unacceptable submissions. Comparisons of average and distributional differences between labeled classes will help determine whether surface-level features such as length, word diversity, or structural completeness are predictive of quality. For example, if unacceptable submissions consistently include shorter analysis sections or fewer references to performance benchmarks, these patterns may inform the design of engineered features.

We will also analyze metadata distributions across programs, schools, degree levels, and years. Frequency tables and bar charts will be used to identify whether certain schools or program types are overrepresented in the labeled subset. Temporal visualizations will help detect potential shifts in submission patterns or labeling tendencies over time. This will inform stratified validation strategies and guard against confounding institutional or temporal effects.

The second stage of EDA will focus on textual and linguistic exploration. Basic natural language processing techniques will be applied to examine vocabulary distributions across PLOs and ROAR sections. These insights will guide feature construction for classification models.

We will also conduct exploratory semantic similarity analysis between PLO statements and their associated ROAR assessment descriptions. Initial similarity scores using baseline embeddings will help establish whether measurable alignment differences exist between flagged and non-flagged submissions. If semantic similarity appears weakly associated with quality labels, this will justify deeper modeling approaches using contextual embeddings or transformer-based methods.

Where quantitative results can be extracted reliably, we will examine distributions of reported percentages, benchmarks, and attainment thresholds. Summary statistics and density plots will help determine whether unacceptable submissions are associated with missing numeric targets, extreme values, or inconsistent reporting formats. If numeric presence itself is predictive, binary indicators may be engineered as features.

Additionally, we will assess section-level completeness by detecting the presence or absence of expected components (methods, results, analysis, follow-up). Simple rule-based checks will help quantify structural compliance rates. This analysis will inform the development of automated completeness flags described in earlier project objectives.

Finally, dimensionality reduction techniques such as Principal Component Analysis (PCA) or visualization of embedding spaces may be used to explore clustering patterns among PLO-ROAR pairs. If acceptable and unacceptable submissions show separable clusters in embedding space, this would provide preliminary evidence supporting classification feasibility.

Overall, EDA will inform:

1. Selection of text representations
2. Identification of engineered structural features
3. Validation strategy design
4. Decisions regarding model complexity given the dataset size

By grounding modeling choices in empirical patterns observed during exploratory analysis, this phase will reduce the risk of overfitting, improve interpretability, and ensure that subsequent modeling decisions are aligned with the actual characteristics of the data.

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3. Data Science Pipeline Design Plan

3.1 Problem Formulation

We formulate the analysis of ROAR submissions as a supervised binary classification problem. The primary data science objective is to develop a predictive model that identifies “Unacceptable” or “At-Risk” ROAR submissions, distinguishing them from “Acceptable” ones based on structural or semantic features.

The problem is decomposed into:

1. Structure and Completeness Verification:

   The first sub-task is to parse PDF documents to extract key narrative fields (Program Learning Outcomes, Methods, Results, Actions). We frame this as a compliance check where the system must identify missing fields, formatting inconsistencies, or insufficient length. The goal is to flag submissions that fail to meet basic reporting standards, regardless of their semantic content.

2. Semantic Quality Classification:

   The second sub-task addresses the semantic coherence of the report. We define this as a classification problem where the model predicts the probability of a submission being "Unacceptable" based on the alignment between its sections. Specifically, we aim to detect misalignments through results that do not correspond to the stated methods or vague language that characterizes low-quality submissions.

The ultimate goal is decision support rather than fully automated grading; the system output will be framed as a set of interpretable flags. These flags serve to prioritize manual review, directing staff attention to submissions that are statistically likely to require revision.

3.2 Planned Data Preprocessing and Feature Engineering

1. Text cleaning:

   Remove headers, formatting artifacts.

   Normalize whitespace.

   Section segmentation: Split each ROAR into Methods, Results, Improvements, and Follow-up sections.

2. Feature Engineering:
   A. Semantic Alignment Features:

      To capture contextual semantic meaning, we will use pretrained transformer-based models such as BERT to generate contextual embeddings for PLO statements and ROAR sections. Unlike TF-IDF, which captures surface-level lexical patterns, BERT

embeddings encode contextual word usage and sentence-level meaning. Cosine similarity between contextual embeddings will serve as a feature measuring conceptual alignment between PLOs and assessment descriptions.

B. Assessment Quality Indicators:

Binary Indicators (numeric threshold, percentage, rubric-related terms, direct assessment, re-assessment).

Quantitative Features (number of numeric expressions, count of action verbs, lexical diversity, section length balance).

C. Logical Structure Features:

Presence of causal markers (“based on”, “as a result”).

Similarity between the result section and the improvement section.

3.3 Modeling Strategy

Baseline Model:

1. TF-IDF + Logistic Regression or TF-IDF + Linear SVM.
2. TF-IDF captures discriminative lexical patterns across ROAR submissions.
3. Linear classifiers are well-suited for high-dimensional sparse text data and perform reliably with small labeled datasets.
4. Provides a stable, interpretable benchmark to assess whether surface-level language differences can distinguish acceptable from problematic ROARs.

Intermediate Model:

1. Pretrained Sentence Embeddings + Supervised Classifier (e.g., Logistic Regression or Gradient Boosting).
2. Embeddings capture deeper semantic relationships between PLO statements and ROAR sections.
3. Cosine similarity features will measure semantic alignment between PLOs and assessment evidence.

Transformer-Based Model:

As an advanced modeling approach, we will evaluate transformer-based architectures such as BERT. BERT enables contextualized language representations that capture deep semantic relationships across report sections. We will explore two approaches:

1. Feature-based approach: Use frozen BERT embeddings as input features to a lightweight classifier.
2. Fine-tuning approach: Fine-tune a small BERT model directly on the labeled ROAR dataset for binary classification. The currently labeled dataset consists of 39 flagged ROARs; additional labeled examples are expected as the project progresses. We will carefully evaluate whether fine-tuning provides measurable gains over frozen embeddings and linear baselines, and apply regularization techniques to mitigate overfitting.

3.4 Validation and Evaluation Plan

1. Stratified k-fold cross-validation:

   The labeled dataset will be split into five folds while preserving the class distribution (acceptable vs. problematic ROARs) in each fold. In each iteration, four folds will be used for training and one fold for validation. Performance will be averaged across folds to obtain stable and reliable estimates. This approach is appropriate given the relatively small dataset size and helps reduce variance caused by a single train/test split. Because the labeled dataset is imbalanced (13 unacceptable vs. 26 acceptable submissions), we will evaluate class-weighted loss functions and threshold calibration to ensure adequate sensitivity to problematic ROARs.

2. Evaluation metrics:
   A. Recall: Measures the proportion of problematic ROARs correctly identified.
   B. Precision: Measures the proportion of flagged ROARs that are truly problematic, controlling unnecessary manual review burden.
   C. F1-score: Harmonic mean of precision and recall, providing a balanced assessment of model performance.
   D. ROC-AUC: Evaluates the model's ability to discriminate between acceptable and problematic ROARs across classification thresholds.

Because transformer-based models have high capacity, cross-validation performance comparisons between baseline linear models and BERT-based models will determine whether the additional model complexity provides meaningful predictive improvements.

3.5 Tools and Technologies

1. Programming Environment: Python, server environment provided by OIE.
2. Data processing and Analysis: pandas and NumPy, will use custom scripts for segmentation of ROARs.
3. Pretrained transformer models (e.g., BERT, DistilBERT, or Sentence-BERT variants) accessed via Hugging Face Transformers for contextual embedding generation and optional fine-tuning.

4. Expected Next Steps

Develop a PDF/Word document analysis system:

1. Use Python libraries to parse the Nuventive PDF reports and segment them into Program Learning Outcomes (PLOs), Assessment Methods, Results, Analysis, and Improvement Plans.
2. Match the extracted PLOs with their associated degree programs. Having this mapping of PLOs to degree programs will allow us to identify low and high-quality PLOs. We hypothesize that low-quality PLOs will have a higher number of poor ROARs due to a lack of clarity.

Create a benchmark for what defines a “good” PLO:

1. Use Rice’s secure Gemini chat to perform in-context learning (provide Gemini prompts with a few examples so the LLM can familiarize itself with the data). The models will then understand the distinguishing semantic features between good and bad PLOs.
2. Select semantically significant verbs (i.e., research) that appear in good PLOs to curate a set of modeling features.

Implement a ROAR flagging system and prioritize ROARs with many of the following flags for review:

1. Basic rule violation: submission is flagged due to missing performance targets, copied content from a previous year’s submission, very little content, or other obvious structural problems.
2. Semantic issues: submission is flagged due to passive verbiage, multiple goals, or vague wording that makes it difficult to logically map assessment results to the measurement method.

Use Rice’s Secure Azure AI Foundry to fine-tune/familiarize an LLM with ROAR quality assessments:

1. Deploy provided models from Azure AI Foundry.
2. Fine-tune small transformer models on the existing labeled data (39+ flagged roars) to improve the accuracy of the selected pre-trained model. This should improve the LLM's performance on ROAR quality assessment.

Validate the system, develop an interface, and get stakeholder feedback:

1. After February 26th, the OIE sponsors will provide realistic mock data that we can use as a testing/validation set. We will use this to gain a baseline understanding of our system’s accuracy and flagging ability.
2. We will develop a user interface where OIE reviewers can look at a dashboard of flagged submissions and verify findings.
3. Allow OIE to beta-test the system and evaluate its effectiveness. If proven effective enough, consider scaling the tool for academic departments to self-screen their submissions before finally submitting to the OIE.